

Quality Improvement Project Template

CBME Competency Development for MBBS Community Medicine Posting

QUALITY IMPROVEMENT (QI) FRAMEWORK

Why QI Matters in Community Diabetes Care

- **Outcomes:** Improve glycemic control and complication rates
 - **Systems:** Strengthen referral and follow-up processes
 - **Education:** Enhance patient and healthcare provider knowledge
 - **Equity:** Reduce disparities in diabetes care access
 - **Sustainability:** Build long-term community health capacity
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QI PROJECT SELECTION CRITERIA

SMART Criteria for Project Selection

- **Specific:** Clear, well-defined problem and objective
- **Measurable:** Quantifiable outcomes and process indicators
- **Achievable:** Realistic given resources and time available
- **Relevant:** Important to patient care and institution
- **Time-bound:** Complete within posting duration

Potential QI Projects

Primary Care Level Projects

1. **Diabetes Screening Quality Improvement**

- Problem: Low screening completion rates in rural areas
- Intervention: Mobile screening camp with improved patient flow
- Outcome: 80% screening completion within 30 minutes

2. **Patient Education Effectiveness**

- Problem: High non-adherence to lifestyle modification
- Intervention: Peer-support group establishment
- Outcome: 70% patients reporting improved medication adherence

3. **Referral System Optimization**

- Problem: Lost-to-follow-up rates >30%
- Intervention: Standardized handover protocol with SMS reminders
- Outcome: <10% loss-to-follow-up at 3 months

Hospital Level Projects

1. **Foot Care Service Enhancement**

- Problem: Delayed presentation of diabetic foot complications
- Intervention: Community education + early intervention clinic
- Outcome: 50% reduction in foot amputation rates

2. **QoL Assessment Integration**

- Problem: Limited attention to psychological impact of diabetes
- Intervention: Routine depression screening and referral protocol
- Outcome: 60% identified patients receive appropriate care

QI METHODOLOGY - PDSA CYCLE

PDSA Cycle Stages

P (Plan): Design intervention based on problem analysis

D (Do): Implement intervention on small scale

S (Study): Analyze results and feedback

A (Act): Refine and implement broadly or discontinue if ineffective

Multiple PDSA Cycles

Plan Implementation → Study Results → Act on Findings

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Plan-Do-Study-Act (Repeat with Refinements)

STEP-BY-STEP QI PROJECT DEVELOPMENT

Step 1: Problem Identification & Definition

AIMS Template Structure:

What are we trying to accomplish?

- Increase screening completion rate for diabetes in rural communities

What will it look like when it is accomplished?

- 95% eligible adults screened annually

- Positive feedback from community members

- Integration into routine PHC services

What changes can we make that will result in an improvement?

- Mobile screening units with facilitated access
- Community health education campaigns
- ASHAs coordinating appointment scheduling

What are we measuring to know that a change is an improvement?

- Screening completion percentage
- Patient satisfaction scores
- Cost per screening completed

Step 2: Analysis & Cause-Out Root

5-Whys Technique:

Problem: Only 40% of eligible adults get screened for diabetes annually
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 Why? Screening facility is 5 km away with poor road access
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 Why? No mobile screening services available
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 Why? Limited transportation budget for health department
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 Why? Priority given to curative services over preventive care
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 Solution: Advocate for preventive care budget allocation + mobile clinics

Fishbone Diagram Categories:

- **People:** Provider knowledge, patient awareness, staff training
- **Processes:** Appointment systems, clinic workflow, referral mechanisms
- **Equipment:** Availability of glucometers, maintenance schedules
- **Environment:** Distance to clinic, social stigma, economic barriers
- **Materials:** Informed consent forms, educational materials
- **Policies:** Government guidelines implementation, insurance coverage

Step 3: Intervention Design

Change Package Development:

Intervention Elements:

1. Mobile Screening Camp Schedule (Weekly rotation)
2. Community Awareness Campaign (Local radio + posters)
3. ASHA Coordination System (Door-to-door mobilization)
4. Appointment Recall System (SMS reminders)
5. Patient Feedback Mechanism (Suggestion boxes)

Implementation Checklist:

- ☐ Trained staff for mobile unit operations

- Supplies stocked and validated
- Community leaders informed and supportive
- Emergency response protocol established
- Data collection forms prepared
- Quality control measures in place

Step 4: Measurement & Data Collection

Outcome Measures (Results Focus):

- **Process:** % eligible adults screened
- **Impact:** HbA1c reduction in diagnosed patients
- **Outcome:** Hospitalization rates for diabetic complications

Balancing Measures (Unintended Consequences):

- Patient satisfaction with screening process
- Staff workload and burnout levels
- Cost per patient screened
- No-show rates and appointment wastage

Data Collection Methods:

Quantitative Methods:

- Screening rates (daily/monthly tracking)
- Blood glucose distribution curves
- Referral completion percentages
- Cost analysis per patient screened

Qualitative Methods:

- Patient feedback interviews
- Healthcare provider experience surveys
- Community leader consultations
- Process observation notes

Step 5: PDSA Cycle Implementation

Plan Phase:

1. **Objective Statement:** Develop 3-month plan to increase diabetes screening by 50%
2. **Intervention Details:** Weekly mobile camps + community education
3. **Success Criteria:** >60% screening completion in pilot areas
4. **Data Collection Plan:** Daily screening logs + monthly patient surveys

Do Phase:

1. **Pilot Implementation:** Start with one village, 4-week intervention
2. **Team Responsibilities:** Each student takes specific roles (screening, education, data)

- 3. **Quality Control:** Daily review of glucometer accuracy, consent procedures
- 4. **Contingency Plans:** Backup power, weather delays, equipment failure protocols

Study Phase:

- 1. **Data Analysis:** Daily screening numbers, quality of service metrics
- 2. **Process Review:** Team debriefs to identify successes and challenges
- 3. **Patient Feedback:** Experience surveys and suggestion collection
- 4. **Comparative Analysis:** Compare pilot with baseline performance

Act Phase:

- 1. **Scale-Up Decisions:** Successful interventions expanded to more villages
- 2. **Refinement:** Based on lessons learned from pilot implementation
- 3. **Sustainability Planning:** Training of local health workers for continuation
- 4. **Documentation:** Final project report with recommendations for health system

CBME ASSESSMENT RUBRICS FOR QI PROJECTS

Assessment Criteria for Student QI Projects

1. Project Planning (20 Points)

- Clarity of problem definition and scope
- Appropriateness of methodology selection
- Feasibility assessment completeness
- Resource allocation planning

2. QI Methodology Application (25 Points)

- Proper use of PDSA cycles
- Data collection rigor and accuracy
- Statistical analysis appropriateness
- Valid interpretation of results

3. Intervention Effectiveness (25 Points)

- Measurable improvement achievement
- Balancing measures consideration
- Sustainability planning adequacy
- Cost-effectiveness demonstration

4. Communication & Leadership (15 Points)

- Team collaboration effectiveness
- Stakeholder engagement quality
- Presentation clarity and impact
- Leadership and problem-solving skills

5. Overall Impact (15 Points)

- Health outcomes improvement
- System-level changes initiated
- Community engagement achieved
- Scalability and generalization potential

SAMPLE QI PROJECT TOPIC: DIABETES SCREENING IMPROVEMENT

Project Title

"Improving Diabetes Screening Coverage in Rural Karnataka Through Mobile Health Camps"

Problem Statement

In our assigned rural community, only 35% of adults aged 30-60 years complete annual diabetes screening, below the national target of 60%. This leads to delayed diagnosis and higher complication rates.

Root Cause Analysis

- **Access issues:** 3-5 km distance to PHC, poor roads during monsoons
- **Awareness gaps:** Limited understanding of asymptomatic nature of diabetes
- **Perception barriers:** "Screening only for sick people" stigma
- **Supply constraints:** Limited testing supplies and staff time

Intervention Design

1. **Mobile Diabetes Van:** Equipped with glucometer, educational materials, trained staff
2. **Weekly Schedule:** Rotation through 8 villages in 2-month period
3. **Community Mobilization:** ASHA coordination + local announcements
4. **Screening Package:** FBG + education + follow-up appointment scheduling
5. **Quality Standards:** <15 minutes per patient, >95% consent rates

Expected Outcomes

- **Process:** 70-80% screening completion in target villages
- **Impact:** Early detection of 15-20% new diabetes cases
- **Outcome:** Improved glycemic control through timely management
- **Sustainability:** Model that can be adopted by local PHCs

STUDENT QI PROJECT EVALUATION RUBRIC

For Faculty Assessment

Competency	Needs Development	Meets Expectations	Exceeds Expectations
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Competency	Needs Development	Meets Expectations	Exceeds Expectations
Problem Identification	Problem vaguely stated	Clear problem with data	Data-driven analysis with root cause
Method Selection	Basic PDSA attempted	Full QI methodology used	Innovative adaptations made
Data Analysis	Basic statistics	Proper statistical tests	Advanced analysis with modeling
Intervention Design	Simple changes	Theory-informed approach	Evidence-based innovative design
Results Communication	Basic report	Clear presentation	Impactful dissemination beyond project

Self-Assessment Questions

- Did I clearly define measurable outcomes?
- Did I collect baseline and follow-up data?
- Did I consider balancing measures?
- Did I involve stakeholders appropriately?
- Did I create a sustainable solution?

PROJECT TIMELINE

Week 1-2: Exploration & Planning

- Literature review on diabetes screening QI
- Stakeholder interviews and community assessment
- Baseline data collection and problem analysis
- Project proposal development and approval

Week 3-6: Intervention Implementation

- PDSA cycle 1: Small-scale pilot implementation
- Data collection and real-time monitoring
- Mid-course corrections based on initial results
- Stakeholder feedback incorporation

Week 7-8: Analysis & Refinement

- PDSA cycle 2: Refined intervention implementation
- Comprehensive data analysis and evaluation
- Community impact assessment
- Sustainability planning development

Week 9-10: Dissemination & Scaling

- Results presentation to stakeholders
 - Project report and recommendations
 - Knowledge transfer to local health system
 - Handover for continued implementation
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RESOURCE REQUIREMENTS FOR QI PROJECTS

Human Resources

- **Student Team:** 4-6 MBBS students with faculty supervision
- **Local Support:** ASHA workers, ANMs for community mobilization
- **Technical Support:** Data analyst, statistical consultant if needed
- **Community Team:** Local leaders, school teachers for awareness

Material Resources

- **Data Collection:** Survey forms, data entry software, storage solutions
- **Analysis Tools:** Excel/SPSS, data visualization software
- **Communication:** Projector for presentations, printing for reports
- **Implementation:** Equipment for surveys, incentives for participants

Budget Considerations

- **Equipment:** ₹2,000-5,000 (forms, stationery, software subscriptions)
 - **Transportation:** ₹3,000-5,000 (field visits to project sites)
 - **Communication:** ₹1,000-2,000 (printing, refreshments)
 - **Miscellaneous:** ₹2,000 (emergency funds, contingencies)
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STUDENT SKILL DEVELOPMENT THROUGH QI

Core Competencies Developed

- **Critical Thinking:** Problem analysis, solution design
- **Research Skills:** Literature review, evidence-based practice
- **Data Management:** Collection, analysis, interpretation
- **Communication:** Presentation, stakeholder engagement
- **Leadership:** Team management, change implementation
- **Systems Thinking:** Healthcare process improvement

CBME Domain Alignment

- **Patient Care:** Improved care delivery processes
 - **Medical Knowledge:** Application of evidence in practice
 - **Practice-Based Learning:** Self-directed improvement
 - **Communication Skills:** Multi-stakeholder collaboration
 - **Professionalism:** Ethical practice, accountability
 - **Systems-Based Practice:** Quality and systems improvement
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QUALITY ASSURANCE STANDARDS

Project Quality Checklist

- ☒ Ethical review and approval obtained
- ☒ Data collection processes validated
- ☒ Intervention fidelity maintained
- ☒ Balancing measures monitored
- ☒ Stakeholder feedback integrated
- ☒ Results accurately reported

Documentation Standards

- **Problem Statement:** Clear, measurable, important
 - **Methodology:** Replicable and appropriate
 - **Data Analysis:** Accurate and unbiased
 - **Conclusions:** Supported by evidence
 - **Recommendations:** Practical and actionable
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IMPLEMENTATION CHALLENGES & SOLUTIONS

Common Student Challenges

- **Time Management:** Solution - Weekly planning and monitoring
- **Data Collection:** Solution - Simple, standardized methods
- **Stakeholder Resistance:** Solution - Early engagement and education
- **Complex Analysis:** Solution - Faculty guidance and software tools
- **Resource Limitations:** Solution - Start small, build on successes

Faculty Support Strategies

- **Mentorship:** Regular check-ins and guidance
 - **Resource Provision:** Templates, tools, statistical support
 - **Skill Building:** Workshop on QI methodologies
 - **Problem-Solving:** Troubleshooting consultation
 - **Quality Review:** Preview of data analysis and reports
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FINAL PROJECT REPORT TEMPLATE

Executive Summary (1 page)

- Problem statement and significance
- Intervention description and implementation
- Key results and outcomes
- Recommendations for practice/policy

Background & Literature Review (2-3 pages)

- Local and national context
- Evidence from literature
- Theoretical framework used

Methodology (3-4 pages)

- Study design and PDSA cycles
- Data collection and analysis methods
- Quality control measures
- Limitations and solutions

Results & Analysis (4-5 pages)

- Quantitative data and statistical analysis
- Qualitative findings and themes
- Process measures and balancing measures
- Subgroup analyses and unexpected findings

Discussion & Recommendations (3-4 pages)

- Interpretation of results
- Comparison with literature
- Strengths and limitations
- Implications for practice
- Policy recommendations
- Future research directions

Appendices

- Survey instruments and protocols
- Raw data tables and figures
- Consent forms and documentation
- Budget and timeline details
- Team member contributions

Quality Improvement Project Template: This comprehensive framework enables MBBS students to conduct meaningful improvement projects that enhance diabetes care systems while developing CBME competencies essential for medical practice.

Faculty Note: Encourage students to select projects that align with their interests and have potential for real impact in the community setting.